RV College of Engineering®

Academic year 2025-2026 (ODD Sem)

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

Date: 07/11/2025	**CIE - 1**	**Max. Marks:(10+50) = 60**
Semester: V	**UG**	**Duration: 2 Hours**
Course Title: Artificial Intelligence and Machine Learning (Common to CSE/ISE/CD/CY)		**Course Code: IS353IA**

Q. No.	QUIZ	M	BT	CO
1.	An empirical test was proposed by Alan Turing which was designed to provide a satisfactory operational definition of Intelligence. List the important capabilities the computer needs to possess to pass Turing test and Total Turing Test.	2	L2	CO1
2.	List the four basic kinds of agent programs that embody the principles underlying almost all intelligent systems.	2	L2	CO1
3.	Decision tree induction algorithms must provide a method for expressing an attribute test condition and its corresponding outcomes for different attribute types. Indicate with the aid of suitable pictorial representation (considering real-world examples) for nominal and ordinal attributes.	2	L3	CO2
4.	Draw the General framework for building a classification model.	2	L2	CO3
5.	Determine the different impurity measures for evaluating a Single Node given below:	2	L3	CO3

Node N_2	Count
Class=0	1
Class=1	5

Q. No.	TEST	M	BT	CO
1a.	Summarize in a tabular form the PEAS description of the task environment for Part-picking robot. Identify and list the dimensions (properties) along which task environment of Part-picking robot can be categorized. Justify your answer	05	L3	CO1
1b.	Draw the schematic diagram indicating Structure of Model-based Reflex Agent and discuss the characteristics considering the Scenario of Automated Taxi. Write the pseudocode for Agent program for Model-based Reflex Agent	05	L2	CO1
2a.	Consider a state space where the start state is number 1 and each state k has two successors: numbers 2k and 2k + 1. i. Draw the portion of the state space for states 1 to 15. ii. Suppose the goal state is 11. List the order in which nodes will be visited for breadth first search, depth-limited search with limit 3, and iterative deepening search. Indicate all the intermediate steps representing the solution.	06	L3	CO2
2b.	Suppose that an agent lives in a grid world of size 5 x 5 (for a total of 25 squares). The agent has two sensors: a GPS sensor, which informs the agent of its current location on the grid, and a camera sensor, which informs the agent of the color on the current square and the four adjacent squares. The agent, at each step, moves left, right, top, bottom. 24 of the 25 squares are safe, and one square (at location 4,3) is dangerous. The current location of the agent is safe.	04	L3	CO2

	a. If the agent is reflex-based, the safe squares are green, and the dangerous square is red, is it possible for this agent to follow a safe strategy that will always avoid the dangerous square? If yes, what is that strategy? b. If the agent is reflex-based, and all squares (safe and dangerous) are green, is it possible for this agent to follow a safe strategy that will always avoid the dangerous square? If yes, what is that strategy?			
3a.	Write the pseudocode for decision tree induction algorithm. Explain the application of decision trees briefly for any real-world use case.	05	L3	CO3
3b.	A model is built to detect whether a patient has a disease. The model was tested on 300 patients. Out of 120 patients who actually had the disease, 90 were correctly diagnosed. The model also incorrectly predicted 60 healthy individuals as diseased. Using the above information, compute the following: i) Confusion matrix ii) Accuracy iii) Error Rate (Assume "Has disease" = Positive class)	05	L3	CO5
4.	Consider the training examples shown in Table 5 for a Binary classification problem. Determine the following measures that can be used to determine the goodness of an attribute test condition: (i) What is the entropy of this collection of training examples with respect to the positive class? (ii) What are the information gains of a1 and a2 relative to these training examples? (iii) For a3, which is a continuous attribute, compute the information gain for every possible split. (iv) What is the best split (among a1, a2, and a3) according to the information gain? *Table 4 : Training Samples for a Binary classification problem*	10 (2+4 +3+1)	L4	CO5

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

5a.	Consider the training examples shown in Table 4 for a Binary classification: (i) What is the best split (between a1 and a2) according to the misclassification error rate? (ii) (ii) What is the best split (between a1 and a2) according to the Gini index?	05 (3+2)	L4	CO5
5b.	Summarize any 5 important characteristics of decision tree induction algorithms.	05	L2	CO2

Course Outcomes: After completing the course, the students will be able to: -	
CO1:	Explain and apply AI and ML algorithms to address various requirements of real-world problems
CO2:	Design and develop AI and ML solutions to benefit society, science, and industry.
CO3:	Use modern tools to create AI and ML solutions.
CO4:	Demonstrate effective communication through team presentations and reports to analyze the impact of AI an ML solutions on society and nature.
CO5:	Conduct performance evaluation, modeling, and validation of AI and ML solutions benefiting lifelong learni